GFS SPECIFICATION

FOUNDATION, ANCHOR BOLTS AND GROUTING FOR ONSHORE LNG

GFS IG.34.85.20-Gen.

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GENERAL FUNCTIONAL STANDARD

PREFACE

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These views are based on the experience acquired during involvement with the design, construction, operation and maintenance of processing units and facilities. Where deemed appropriate GFSs are based on Shell DEPs, international, and industry standards.

The objective is to set the standard for good design and engineering practice to be applied by Shell companies in oil and gas production, oil refining, gas handling, gasification, chemical processing, or any other such facility, and thereby to help achieve maximum technical and economic benefit from standardization.

The information set forth in these publications is provided to Shell companies for their consideration and decision to implement. This is of particular importance where GFSs may not cover every requirement or diversity of condition at each locality.

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All administrative queries should be directed to the Technical Standards Team in Shell GSI.

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1. INTRODUCTION

1.1 SCOPE

This new GFS specifies minimum requirements and gives recommendations for the structural design, detailing and construction of the following:

- shallow individual 'pad' foundations supporting pipe racks, structural frames, pipe supports, equipment, buildings;
- combined foundations supporting multiple pedestals or pipe supports;
- flexible raft foundations;
- light machinery foundations;
- anchor bolt and grouting.

Heavy machinery foundations are specifically excluded from this GFS and are defined as follows:

- foundations supporting equipment having reciprocating or rotary masses with a total weight greater than 2268 kg (5000 lb) or a gross plan area greater than 2.5 m^2 (26.9 ft^2); or
- reciprocating and centrifugal machines greater than 150 kW (200HP) and 375 kW (500 HP) horsepower.

Heavy machinery foundations shall be designed and detailed in accordance with GFS.Machinery.GFS.

This GFS is classified as a TIER 2 document. See (1.3) for definitions of TIER 1/TIER 2.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this GFS is confined to Shell companies and, where necessary, to Contractors nominated by them. Any authorised access to a GFS does not for that reason constitute an authorisation to any documents, data or information to which the GFS may refer.

This GFS is intended for use in facilities related to Onshore LNG. This GFS may also be applied in other similar facilities.

When GFSs are applied, a Management of Change (MOC) process shall be implemented; this is of particular importance when existing facilities are to be modified.

If national and/or local regulations exist in which some of the requirements could be more stringent than in this GFS, the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and, which combination of requirements will be acceptable with regards to the safety, environmental, economic and legal aspects. In all cases, the Contractor shall inform the Principal of any deviation from the requirements of GFS which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned the objective being to obtain agreement to follow this GFS as closely as possible.

1.3 DEFINITIONS

1.3.1 General Definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

TIER 1 packaged equipment solutions are kept evergreen and have a revision cycle.

TIER 2 packaged equipment solutions do not have a revision cycle and updated when needed.

The word **shall** indicates a requirement.

If deviating from a requirement represented as **shall**[1], refer to the Principal for the appropriate deviation process.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

1.3.2 Specific Definitions

Term	Definition
Cement based grout	A type of grout used to fill voids and cracks in pavements, building foundations, and brick and concrete masonry wall assemblies. Cement based grouts are also used between baseplates, soleplates and the foundation; to construct floor toppings or provide flooring underlayment; to place ceramic tiles and to bind preplaced-aggregate concrete.
Epoxy grout	A two-part grout system consisting of epoxy resin and epoxy hardener that is made to have impervious qualities, such as being stain and chemical resistant. Epoxy grout is used to fill joints between tiles, anchors, baseplates, etc.
Grouting	Work carried out to properly fill spaces between concrete surfaces and baseplates of equipment, steel structures or prefabricated concrete elements with mortar in order to achieve the adequate transfer of horizontal and vertical forces.
Light machinery foundation	Foundations supporting rotating equipment including small centrifugal and vertical pumps, engines, turbines and compressors having reciprocating or rotary masses, with a total weight less than 2268 kg (5000 lb) or a gross plan area of less than 2.5m^2 (26.9 ft^2), centrifugal machinery less than 375 kW (500HP) or reciprocating machinery less than 150 kW (200 HP).
Technical Authority	Refer to the DCAF system.

1.3.3 Abbreviations

Term	Definition
ACI	American Concrete Institute
API	American Petroleum Institute
ASD	Allowable Stress Design
ASTM	American Society for Testing and Materials
GFoS	Global Factor of Safety

Term	Definition
HPP	Highest Pavement Point
ITP	Inspection and Test Plan
LRFD	Load Resistance Factor Design
OBE	Operating Basis Earthquake
QA/QC	Quality Assurance/Quality Control
QCR	Quality Control Records
SSE	Safe Shutdown Earthquake

1.4 CROSS-REFERENCES

Where cross-references to other parts of this GFS are made, the referenced section or clause number is shown in brackets (). Other documents referenced by this GFS are listed in (7).

1.5 SUMMARY OF MAIN CHANGES

This is a new GFS.

1.6 COMMENTS ON THIS GFS

Comments on this GFS may be submitted to the Administrator using the same methods as for submitting comments on DEPs:

Shell DEPs Online (Users with access to Shell DEPs Online)	Enter the Shell DEPs Online system at https://www.shelldeps.com Select a GFS and then go to the details screen for that GFS. Click on the “Give feedback” link, fill in the online form and submit.
DEP Feedback System (Users with access to Shell Wide Web)	Enter comments directly in the DEP Feedback System which is accessible from the Technical Standards Portal http://sww.shell.com/standards. Select “Submit DEP Feedback”, fill in the online form and submit.
DEP Standard Form (other users)	Use Standard Form PES 00.00.05.82-Gen. to record feedback and email the form to the Administrator at standards@shell.com.

Feedback that has been registered in the Feedback System by using one of the above options will be reviewed by the GFS Custodian for potential improvements to the GFS.

1.7 DUAL UNITS

This GFS contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this GFS indicates text that is non-normative and is provided as explanation or background information only.

Non-normative text is normally indented slightly to the right of the relevant GFS clause.

1.9 UPDATE POLICY

This GFS will be updated when applied in a project in accordance with the framework for Tier 2 Packaged Equipment Solutions.

2. GENERAL

2.1 PURPOSE OF THIS GFS

This GFS provides the basis for the development of project specifications and supporting documentation for the design and construction of foundations, anchor bolts and grouting for projects based on the National Standards supplemented by the requirements of this GFS and project specific data as specified in the contract documents.

1. Any conflicts or inconsistencies between this GFS, design drawings, and project specifications shall be brought to the attention of the Principal for resolution.

2.2 APPLICABLE CODES AND STANDARDS

This GFS is based on American standards and provides minimum requirements for the structural design of shallow foundations, grouting and anchor bolts.

1. Unless specifically designated by a date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.

2.3 DESIGNER RESPONSIBILITY

1. The Contractor shall have the overall responsibility for the design of the shallow foundation.
2. Any recommendation related to safety, constructability and maintainability shall be detailed by Geotechnical Contractor and is deemed to be part of the scope of work.

3. FOUNDATIONS

3.1 GENERAL

1. Foundations shall be designed so that the net bearing capacities are not exceeded and differential settlements are minimised.
   a. Net bearing capacities shall be based on the results of a geotechnical investigation.
      i. Net bearing capacities shall comply with LFRD or GFoS methodologies as documented in the GFS IG.34.85.23-Gen.
2. Load Resistance Factor Design (LRFD) shall be used for the strength design of the anchor bolts and the concrete elements of the foundation system.
   a. Load factors and combinations of factors shall be in accordance with GFS IG.34.85.02-Gen.
3. The minimum sliding and overturning factors of safety against all serviceability combinations, as documented in GFS IG.34.85.02-Gen., shall be greater than the following:
   a. overturning:
      i. 2.0 under normal operating conditions;
      ii. 1.5 under extreme environmental and OBE;
      iii. 1.2 under abnormal conditions such as blast and SSE.
   b. 1.5 sliding;
   c. 1.1 buoyancy.

4. Sliding resistance shall be developed by friction between footing and soil only.
   a. passive resistance shall be included in the sliding resistance calculation, only if shear keys are provided below the footing;
   b. where incorporated, passive resistance factors shall be as per GFS IG.34.85.23-Gen.
5. Foundations subject to temperature effects shall be designed for any temperature difference that may occur.
6. The ratio of foundation mass to equipment mass for rotary and reciprocating foundations shall be a minimum of 3:1 and 5:1 respectively.

   Dynamic analysis is not required.
7. The eccentricity and application points of loadings shall be considered to correctly simulate structural demands in the foundation.
8. Support of miscellaneous vertical loads of 1100 kg (2500 lb) or less, in paved areas may be made integral with the slab on grade.
   a. Provisions shall be made to distribute such loads by locally thickening the paving slab, as required.

3.1.1 Pad Footings (Rigid)

1. Individual pad foundations shall be proportioned and designed as rigid foundations, using manual methods in accordance with the following equation:
   a. λL shall be ≤ π/4, where:

$$\lambda = \sqrt[4]{\frac{K's\,B}{4.E.I}} \qquad \text{Eqn. 3.1}$$

Where:

K'_s = Coefficient of subgrade reaction (upper bound);

L = Side length of the foundation;

B = Equivalent width of foundation (for square foundations B = L);

E = Youngs Modulus of Concrete in accordance with ACI 318;

I = Uncracked stiffness of foundation ($B.t^3/12$);

t = Foundation slab thickness.

   b. The upper bound coefficient of subgrade reaction shall be used for the determination of slab thickness requirements.
   c. Minimum thickness requirements for pad foundations to enable rigid behaviour shall be as per Appendix A.

3.1.2 Combined Footings (Rigid or Flexible)

1. Combined foundations shall be proportioned and designed as rigid foundations, using manual methods, in accordance with ACI 336.2 provided.
   a. the pedestal or column loads and spacings are relatively uniform and vary by no more than 20 %;
   b. the pedestal or column spacing is less than $1.75/\lambda$ and the cantilever length of the footing is less than $0.875/\lambda$;
   c. the upper bound coefficient of subgrade reaction shall be used for the determination of slab thickness requirements.
2. Where the criterion in (1) is not met, or where the loading geometry is complicated, the footing shall be designed as a flexible footing using the finite element in accordance with (3.1.3).
3. Differential settlements between adjacent columns or adjacent structural supports on combined footings shall be limited to +/- 12 mm (1/2 in).
   a. Where the limits above are exceeded, the slab shall be thickened to increase flexural or some other means shall be used to limit displacements.

3.1.3 Raft Footings (Flexible)

1. Raft footings, by the combined effect of geometry and foundation interaction (slenderness) and/or loading complexity shall be designed as flexible foundations using the finite element method.
   a. Soil springs shall be developed to be commensurate with the expected soil strains resulting from the slab deformation.

3.1.4 Vessel Foundations (Rigid)

1. Footings with face to face dimensions greater than 2.1 m (7 ft) shall be octagonal.
2. Pedestal geometry shall be as follows:
   a. octagonal, where face to face dimension ≥ 1200 mm (4 ft);
   b. the height of the pedestal shall be such to contain the anchor bolts and prevent the casting of bolt systems into the footing.
3. Fully anchored perimeter tension steel shall be provided to transfer the tension loads from the pedestal to the footing.
4. Top and bottom faces of the footing shall be reinforced.

3.2 DETAILING

1. The bottom of the foundation shall be located such that it reaches adequate bearing stratum as per the recommendation of the Geotechnical report.
2. Shallow foundations shall be located 300 mm (12 in) below the assessed frost penetration depth.
3. No conduits, services, pipelines or drainage channels shall be located in a zone bounded by the 45 degrees projected planes downward and outward from the bottom of the foundation.
4. The foundation (slab) thickness shall be the maximum of the following:
   a. 300 mm (12 in), increased in 50 mm (2 in) increments;
   b. the thickness required to develop the pedestal anchorage reinforcement; or,
   c. the thickness required to resist the LRFD shear demand without shear reinforcement.

5. A layer of blinding concrete of 50 mm (2 in) minimum thickness shall be placed under all foundations.
6. Foundations shall be provided with reinforcement to the top and bottom faces.
7. The top of foundation pedestals shall be a minimum 150 mm (6 in) above the Highest Pavement Point (HPP).
    a. minimum pedestal dimensions shall be 300 mm x 300 mm (12 in x 12 in);
    b. pedestals shall be increased in 50 mm (2 in) increments;
    c. pedestals shall be sized so that the anchor bolts do not penetrate the main foundation slab i.e., resistance is within the pedestal;
    d. the pedestal shall be sized so that grouting can be extended 50 mm (2 in) horizontally beyond the base/sole plate;
    e. piers and pedestals shall be reinforced to resist self-straining thermal stresses due to the expansion of base plates, anchor bolts and so forth.
8. Minimum reinforcement limits, design and detailing of pedestals where the height to lateral dimension ratio is three or greater shall be designed as a column in accordance with ACI 318.
    a. all longitudinal reinforcement shall be enclosed by lateral reinforcement that surrounds the vertical bars of the pedestal;
    b. minimum lateral reinforcement shall be provided in accordance with Section 10.6.2.2 of ACI 318;
    c. lateral reinforcement shall be distributed within 100 mm (4 in) of the top of the column or pedestal;
        i. lateral reinforcement shall consist of at least two No.4 or three No.3 bars.
    d. ties shall have a hook on each free end for structures assigned to Seismic Design Categories C, D, E or F.
9. Pedestals of foundation for skirted process vessels shall be provided with a slope and drain.

3.3 MATERIAL

1. Materials for foundations shall conform to the requirement in GFS IG.34.85.31-Gen.

3.4 CONSTRUCTION

1. Construction of foundation shall be as per GFS IG.34.85.31-Gen.
2. Construction process and workmanship shall be in accordance with the project specifications.
3. Excavation, site preparation and backfilling for foundations shall be in accordance with GFS IG.34.85.22-Gen.

4. ANCHOR BOLTS

4.1 GENERAL

1. Chemical, or post-installed type anchors, shall not be used to attach rotating equipment to concrete or where the anchor bolts will be subjected to vibratory loads.
2. Hooked bolts (J bolts) are not permitted.
3. Adhesive/Chemical anchors, post-installed expansion and undercut anchors shall not be used unless noted as follows:
   a. For the attachment of ancillary structures, ladders, or cable trays where approved by the Company;
   b. Chemical anchor bolts used for sustained structural loads shall be seismically qualified as per ACI 355.2;
   c. Chemical anchor bolts that are installed into hardened concrete with an engineered proprietary system shall be subject to the approval of the Company;
   d. Chemical anchor bolts used for sustained structural loads shall not be installed in overhead applications;
   e. Expansion anchors shall not be used where structural tension or moment loads are to be resisted.
4. When exposed to temperatures ≥ 60 °C (140 °F), epoxy anchor bolts shall not be used.
5. Minimum design service temperature shall be as per the project specifications.

4.2 DESIGN AND DETAILING

1. For foundations supporting vibrating equipment, anchor bolts shall be installed using sleeves.
2. Cast-in anchor bolts shall be positioned using templates securely tied to local reinforcement to avoid displacement during concreting operations.
3. Anchorage design and construction shall be designed as ductile connections, for Seismic Design Category (SDC) C, D, E, or F, as per Section 17.2.3.43 (a) of ACI 318.
4. Load factors and loading combinations shall be as per GFS IG.34.85.02-Gen.
5. The anchoring system including the anchor bolts and concrete anchorage shall be designed using the LRFD methodology.
6. Anchor bolt sleeves or anchor chairs (bolt boxes) shall be provided as per the requirements of ACI 318.
   a. sleeves and chairs shall be detailed to accommodate the pretension and ‘stretch length’ requirements as per ACI 318 and shall be a minimum of 150 mm (6 in);
   b. the inner diameter of the sleeve shall be at least twice the diameter of the anchor bolts.
7. Anchor bolts subject to uplift or vibration shall be locked with a washer and two nuts as per the requirements of Table 4.3.
8. Concrete pockets shall be provided where anchor bolts are cast-in-place and where a template is not provided to allow for minor field adjustments (< 6mm (1/4 in)).
9. Anchor bolt spacing, concrete edge distance and embedment shall be as per the requirements in Table 4.1.
10. Minimum edge distances for post-installed anchors shall be as per Section 17.7 ACI 318.

11. All threads shall be protected by greasing and wrapping of the threads with a suitable material.

Table 4.1 Anchor bolt placement requirements

			Without sleeve	With sleeve
Concrete edge distance	Machine foundation	U	Max [$4d_a$: 155 mm (6 in)]	Max [$4d_a$: 155 mm (6 in)] + $0.5(d_s - d_a)$
		T	Max [$6d_a$: 155 mm (6 in)]	Max [$6d_a$: 155 mm (6 in)] + $0.5(d_s - d_a)$
	Structural foundation	U	Max [$4d_a$: 115 mm (4.5 in)]	Max [$4d_a$: 115 mm (4.5 in)] + $0.5(d_s - d_a)$
		T	Max [$6d_a$: 115 mm (4.5 in)]	Max [$6d_a$: 115 mm (4.5 in)] + $0.5(d_s – d_a)$
	All foundations	U	$4d_a$	$4d_a + 0.5(d_s – d_a)$
		T	$6d_a$	$6d_a + 0.5(d_s – d_a)$
Embedment	All foundations	$12d_a$		For $d_a \geq 25$ mm (1 in) : Max[$12d_a$: Sleeve length + $6d_a$] For $d_a < 25$ mm (1 in): Max[$12d_a$: Sleeve length + 155 mm (6 in)]
d_a = anchor bolt diameter; d_s = anchor sleeve diameter; U = Un-torqued; T = Torqued				

4.3 MATERIAL

1. Anchor bolt material shall be in accordance with Table 4.2:

Table 4.2 Anchor bolt material requirement

Type	Materials standards
Bolts	ASTM F1554, ASTM A307, Grade A Heavy hex
Washer	ASTM F436 / F436M
Heavy Hex Nut / Thread	ASTM A563 Grade A / Grade C / Grade D

2. The components specified in Table 4.2 shall be mechanically galvanised in accordance with ASTM B695, Class 55.
   a. Where approved by the Company, hot-dip galvanising as per ASTM F2329 may be used.
      i. Nuts shall be re-tapped after hot-dip galvanising.
3. Maximum service temperature for long-term use of hot-dip galvanised anchors shall be less than 199 °C (390 °F).

4.4 CONSTRUCTION CRITERIA

1. Tolerances for anchor bolts shall be as per ACI 117.
2. Anchor bolts shall be fully tightened to the specified tension after grouting of base plates.
3. Anchor bolts and sleeves shall be positioned and secured prior to placing concrete.
4. Prior to grouting of anchor bolt sleeves; it shall be verified that sleeves are clean, dry and have been filled with a non-bonding moldable material to exclude grout from the anchor bolt sleeves.
5. The anchor bolt threads shall be covered with duct tape, foam pipe insulation or other suitable materials to keep them clean and to prevent any damage.
6. Welding of anchor bolts to the reinforcing bars shall not be permitted.
7. Anchor bolt sleeves shall be capped and/or filled with non-bonding moldable material to keep out water, concrete, and debris during the placement of concrete and grout.
8. Wet dipping (placement after concrete pouring) of anchor bolts shall not be permitted.
9. Provision of sleeves and nuts shall be in accordance with Table 4.3.

Table 4.3 Sleeves and nuts

Equipment/Units	Without Sleeves	With Sleeves	Double Nut
All structures, frames, racks, pipe supports, ancillary structures, field assembled units, equipment other than those noted below	X		
Modules and skids	X		
Guyed stack, derricks and communications towers	X		
Prefabricated equipment (boilers, heaters etc.)		X	
Vertical vessels		X	X
Sphere		X	X
Stack, freestanding		X	X
Light machinery	X		X
Heavy machinery		X	X

5. GROUTING

5.1 GENERAL

1. Grout shall be applied between the concrete foundation/pedestals and the baseplates of all steel structures, stationary equipment, rotating and reciprocating equipment and as required per the project drawings or specifications.
2. Grout shall have a design compressive strength greater than or equal to the strength of the foundation concrete.
3. For the installation of grout for rotating and other foundations defined by Principal as critical, Part V of API RP 686 shall be supplemented by the requirements of this GFS.

5.2 DESIGN AND DETAILING

1. A layered combination of non-shrink cement based and epoxy grout shall be used for machinery with large baseplates that have structural webs deeper than 230 mm (9 in), in accordance with the following criteria:
   a. the first layer (which may be composed of more than a single pour) shall be general purpose epoxy grout;
   b. the first layer shall be poured to a level that is 25 mm (1 in) above the bottom of the internal baseplate stiffeners;
   c. the second layer shall not be poured until the first layer has set;
      i. the second layer (which may be composed of more than a single pour) shall be non-shrink cement based grout.
      ii. the second layer shall be poured to a level that is approximately 50 mm (2 in) from the top of the baseplate decking;
   d. the third layer (final layer) shall not be poured until the second layer has set;
      i. the third layer (final layer) shall be general purpose epoxy grout and shall be poured to the top of the baseplate.
2. Joints in grout shall be located as shown in the drawings or as recommended by the grout manufacturer.
3. Grout manufacturer recommendations for grouting shall be considered in grout placement.

5.2.1 Dimensions

1. Unless specified by drawings, the minimum grout thickness shall be a minimum 25 mm (1 in) with a maximum 50 mm (2 in) single pour layer.
2. Expansion joints in epoxy grout shall be across the entire width of the foundation and spaced at a maximum of 1.4 m (48 in).
3. Forms for epoxy grout shall be at the edge of the foundation and shall not be farther than 152 mm (6 in) from the baseplate perimeter.
4. There shall be one 6 mm (1/4 in) minimum diameter vent hole located at every point where air may be trapped by flowing grout.

5.3 MATERIAL

1. Grout types shall be in accordance with Table 5.1 or as per Manufacturer/ Supplier requirements, whichever is stringent.
2. The water used for mixing water, ice, curing, or any other function relating to the placement of concrete shall be in accordance with ACI 318.

Table 5.1 Grout types

Criteria considered	Non-shrink cement based grout	Epoxy grout
Driver horsepower	Less than 375 kW (500 HP)	More than or equal to 375 kW (500 HP)
Rotating equipment speeds	3600 RPM or less	More than 3600 RPM
Combined weight of (machine, driver, baseplate)	Up to 2268 kg (5000 lbs)	More than 2268 kg (5000 lbs)
All reciprocating machinery with driver rating	Less than 37 kW (50 HP)	More than 37 kW (50 HP)
Other Foundations	Non-dynamic foundation	

3. Epoxy grout may be used at locations where exposure to chemical attack from regular operations is possible.

   For details refer to GFS IG.34.85.25-Gen.

5.3.1 Non-Shrink Cement based Grout

1. Non-shrink grouts shall be in accordance with ASTM C1107/C1107M.
   a. metal-oxidising or gypsum-forming non-shrink grout is not permitted;
   b. grout shall not contain metallic substances, aluminium powder, measurable amounts of water soluble chlorides or other substances that may be potentially harmful to concrete or steel reinforcement.
2. No admixtures, unless specified in the contract documents, shall be added to the grout without the approval of the Manufacturer.
3. Mixing water shall be potable and free of oils, acids, alkalines, organics and other deleterious materials.
4. Grout shall have a minimum compressive strength (f’c) of 35 MPa (5100 psi) at 7 days and 40 MPa (6000 psi) at 28 days when tested in accordance with ASTM C 109.
5. Grout shall have a minimum working time of 90 minutes.

5.3.2 Epoxy Grout

1. Compressive strength testing of epoxy grouts shall be in accordance with ASTM C579, Method B.
2. Epoxy grouts for equipment or machine baseplates shall be supplied as specified in the design drawings and/or specification and meet the following chemical and physical properties:
   a. non-rusting or staining;
   b. non-shrink dimensional stability;
   c. flowable placing consistency and full self-levelling characteristics;
   d. non-generation of expansive forces during setting;
   e. non-continuous confinement required during setting period;
   f. unaffected by exposure to the service temperatures;
   g. minimum compressive strength (ASTM C579, Method B modified): 80 MPa (12000 psi) at 7 days;
   h. grout shall have a minimum working time of 90 minutes at 21 ^{o}C (70 ^{o}F);
   i. maximum linear shrinkage (ASTM C531): 0.080 percent;
   j. maximum coefficient of thermal expansion (ASTM C531): 54 x 10^{-6} mm/mm/°C (30 x 10^{-6} inch/inch/°F);
   k. maximum creep (ASTM C1181): 5 x 10^{-6} mm/mm (0.005 in/in), tested at 20 °C (70 °F) and 60 °C (140 °F) with 2.8 MPa (400 psi) applied vertical load;
   l. minimum bond strength of epoxy grout to concrete (ASTM C882/C882M): 14 MPa (2000 psi);
   m. grout shall be resistant to chemical attack of the materials that may continuously drip on the surface.

5.4 CONSTRUCTION CRITERIA

5.4.1 General

1. All materials shall be:
   a. delivered to the job site in original unopened packages;
   b. stored in accordance with the Manufacturer recommendations.
2. All grouting materials and tools shall be stored in a clean, dry and organised manner.
3. Non-shrink, cement based grout and cement intended for use in dry pack shall be in sound dry bags.
   a. Any material that becomes damp or otherwise defective shall be immediately removed from the site.
4. The grout Manufacturer/Supplier shall provide a technical representative who will be on-site for the duration of grouting works to review the following:
   a. the grouting set-up procedures;
   b. preparation of surfaces to be in contact with grout;
   c. material and installation of formwork;
   d. mixing, placement, and curing of grout.

5.4.2 Preparation

1. Foundation preparations shall be in accordance with the Manufacturer written instructions and Section 3.6.2 of API RP 686.
2. The surface of the foundation under the baseplate shall be chipped and brushed to the level of sound fractured, coarse aggregate.
3. Minimum 25 mm (1 in) of the top of concrete shall be removed.
4. Laitance and oil-damaged concrete shall be removed prior to grouting;
    a. Removal of laitance followed by brush hammering alone shall not be acceptable for grout preparation.
5. All dust and loose particles shall be removed with clean, dry, oil-free compressed air after cleaning.
    a. Foundation surface shall be protected from contamination by applying protective sheeting.
6. Unless the grout Manufacturer specifies otherwise, concrete surfaces in contact with the grout shall be saturated with clean water before grout placement for 24 hours (saturated surface dry quality).
7. The surface of the concrete shall be damp but free from standing water.
8. All cement based grout mortar shall be machine mixed to an even consistency.
9. Under no circumstances shall hand mixing be employed.
10. Anchor bolt exposed threads shall be wrapped with duct tape to prevent adherence of epoxy grout.
11. Exposed cement based grout surfaces shall be coated (with an epoxy based product) if future contamination with lubricants is possible.
12. Shims under steel structures and equipment such as columns, vessels, heat exchangers shall be of a machined type.

    Machined type shims need the approval of the equipment Supplier.
13. The following shall apply for shims that will remain in place after grouting:
    a. the shim shall be embedded in a mortar bed so that the top of the shim is level in all planes;
    b. shim plates shall be installed so that they are fully embedded in and surrounded by the grout to be installed later;
        i. the minimum cover to shim sides shall be 50 mm (2 in);
    c. load per shim pack shall not exceed 7 $N.mm^{-2}$ (9.9 psi) under the worst loading condition;
        i. In this case the grout shall be considered as a filling material only when calculating the aforementioned bearing stress.
    d. shim plates are to receive a 6 mm (1/4 in) radius rounded corner;
    e. Wedges shall not remain as a permanent part of the foundation.
        i. Therefore, wedges shall be arranged so that they can be removed and the space backfilled with grout after the initial set and cure of the baseplate grout.

        Use of wedges is restricted to minor steel structures, supports.
14. Levelling screws or counter nuts for machinery shall follow the requirements of API RP 686.

15. For structures and equipment where levelling screws or counter nuts are to remain tight to the baseplate the following shall apply:
    a. the screws or nuts shall be fully embedded in and surrounded by the grout (a mortar bed) so that the baseplate is level in all planes;
        i. the minimum cover to exterior surface of the imbedded screw or nut sides shall be 50 mm (2 in).
    b. the number of screws/level pads selected is not to exceed 7 $N.mm^{-2}$ (9.9 psi) under the worst loading condition to the concrete base or the baseplate;
        i. the grout shall be considered as a filling material only when calculating the aforementioned bearing stress.
    c. the nuts are sized not to exceed 7 $N.mm^{-2}$ (9.9 psi) under the worst loading condition for the baseplate or the capacity of the supporting bolt.
        i. the grout shall be considered as a filling material only when calculating the aforementioned bearing stress.

5.4.3 Formwork

1. Foundations shall be suitably prepared before installation of formwork for grout.
2. All forms shall be built of materials with adequate strength, securely anchored and shored to resist grout placement forces.
3. Formwork for epoxy grout shall be constructed to be liquid-tight with adequate strength, rigidity and dimension to permit epoxy grout placement.
    a. the sealant at the interface between forms and between the forms and concrete foundation shall be per the grout Manufacturer’s recommendations.
4. Forms for cement based grout shall be tight against all surfaces.
5. Joints shall be sealed with tape with the application of form oil or heavy wax for easy form removal.
6. A bond breaker consistent with the Manufacturer’s recommendations shall be applied to the surface of the formwork in contact with the grout.

5.4.4 Joints

1. Joints in grout shall be as shown in the drawings or as recommended by the grout Manufacturer.
2. Expansion joints in epoxy grout shall be across the entire width of the foundation and spaced at a maximum of 1.4 m (48 in).

5.4.5 Environment Weather Considerations

1. Hot or cold weather procedures, as specified by the Manufacturer/Supplier, shall be followed if grouting is to be done when ambient temperatures are not between the following:
    a. 5 °C and 32 °C for cement based grout;
    b. 18 °C and 35 °C for epoxy grout.
2. For extreme weather conditions, grouting shall be designed, executed and tested in accordance with ACI 305R and ACI 306R.

5.4.6 Placement

1. No admixtures, unless specified in the contract documents, shall be added to the grout without the approval of the Manufacturer.
2. Re-tempering of cement based grout by adding water after stiffening shall not be permitted.
3. Grout shall fill all voids between the baseplate and the foundation.
    a. Grout shall have a full contact surface.
4. If anchor bolt sleeves are present, it shall be ensured that sleeves around anchor bolts are, in all cases, kept free from freezing water or snow.
5. If the sleeves are solely for alignment purposes, the anchor bolt sleeves shall be completely filled with grout before placing grout under base and ring plates.
6. Sleeves that allow for pretensioning of the anchor bolt shall be filled with a self levelling sealant as per the drawings before placing grout under the base and ring plates.
7. Cementitious grout may be flowed, or pumped into place as permitted by the Manufacturer’s written instructions for the specific grout being installed.
8. Dry pack may only be used with the approval of the Company.
9. All grout shall be placed in only one direction through drilled vent holes (minimum 6 mm in diameter) in baseplates provided in advance by the Manufacturer.
10. In general, grouting shall be executed over the shortest dimension of the baseplate.
11. The placing of sand-cement grout mortar shall commence not later than 15 minutes after completion of the mixing.
    a. No grout mortar shall be used that has not been placed after 30 minutes from completion of mixing.
12. Re-mixing or knocking up of grout mix shall not be allowed.
13. Top surface of grout shall have a trowel or broom finish per the drawings and specifications.

5.4.7 Curing

1. Grout shall be cured in accordance with the Manufacturer's specification.
2. Cement based grout shall be kept moist during the first 7 days after placing.
    a. Cement based grout shall be protected from sunshine and drying out with a protective covering.
3. After hardening, wedges shall be removed and the resulting voids filled with grout.
4. Exposed expansion joints shall be sealed with the grout Manufacturer’s recommended sealant.

5.4.8 Disposal

1. For epoxy grouts all materials shall be converted to solid waste for proper disposal.
2. The procedure for making solid waste in any type of hardener container is as follows:
   a. mix the hardener with the resin;

      This mixture is considered inert.

   b. pour excess hardener/resin mixture into the hardener container;
   c. close container and shake until excess hardener mixes with excess hardener/resin mixture;
   d. this final mixture in the hardener container is considered inert for proper solid waste disposal.

6. QUALITY AND INSPECTION PLAN

6.1 GENERAL

1. Written Quality Plans and procedures shall be submitted to the Principal for review and approval.
   a. These documents shall provide details of how compliance with the requirements of this GFS, drawings and specifications are to be achieved.
2. Manufacturer's data sheets and certificates, for the materials covered under the scope of this GFS, shall be submitted to the Principal for review and comment.
   a. The submittals shall include the product’s compressive strength, dimensional stability, and other requirements of this GFS.
3. The quality system shall be based on ISO 9000 quality system requirements.
4. Quality plans shall address all aspects related to local conditions, such as climatic conditions, back up facilities, spare parts, transport possibilities, storage facilities, quarries, local Manufacturer/Suppliers, test facilities (field laboratory).

6.2 INSPECTION TESTING AND REPORTING

1. Inspection and Test Plans (ITPs) identifying the tests, frequencies, rejection criteria and responsibilities shall be prepared as part of the quality plan.
2. A reporting system of Quality Control Records (QCRs) shall be part of the quality plan.
3. The reporting system shall record all results obtained during inspection, testing, transporting, production, supplying, placing/pouring, finishing, curing, striking.
4. Weather conditions shall also be recorded in the quality records.
5. The proposed ITPs and QCRs shall be included in the Quality Plan and shall cover the following other items:
   a. setting out;
   b. anchor bolts;
   c. concrete and grout strength;
   d. inserts and other embedment(s);
   e. formwork (including loading capacity, if required);
   f. concrete and grout placing;
   g. workability of concrete and grout;
   h. curing;
   i. removal of formwork (take loading capacity into account);
   j. joint preparations - defects.
6. Standard Forms of Inspection and Test Plans shall be used as a basis for the development of the ITPs and QCRs supplemented by the requirements of the "Specifications" for the work to be executed.
7. After completion, rebound and/or ultrasonic measurements shall be taken to control and verify the concrete quality against the compliance tests and required concrete quality if there is any doubt about the inspection and testing records.
8. Any deviation shall be reported, including the measurements to be taken.
9. Completed grouting checklists found in Chapter Part V of API RP 686 shall be submitted before starting to place epoxy grout.
10. The Principal shall be permitted to make inspections at any time at the source of supply of materials, at the place of preparation of materials, at the mixing plant if ready mixed concrete is used and during execution of all concrete work.

7. REFERENCES

In this GFS, reference is made to the following publications:

NOTES:
1. Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.
2. The referenced external standards are available to Shell staff on the SWW (Shell Wide Web) at http://sww.shell.com/standards/.

7.1 SHELL STANDARDS

SHELL STANDARDS

Procedure for packaged equipment solutions publications	PES 00.00.00.31-Gen.
Packaged equipment solutions feedback form	PES 00.00.05.82-Gen.
Design & Construction of Reinforced Concrete Structures for Onshore LNG Facilities	GFS IG.34.85.31-Gen.
Site Preparation and Earthworks for Onshore LNG Facilities	GFS IG.34.85.22-Gen.
Loading Specification for Onshore LNG Facilities	GFS IG.34.85.02-Gen.
Geotechnical Design of Shallow Foundations for Onshore LNG Facilities	GFS IG.34.85.23-Gen.
General Functional Standard for Heavy Machinery Foundations for Onshore LNG	GFS IG.34.85.25-Gen.

7.2 INDUSTRY STANDARDS

AMERICAN STANDARDS

Specification for Tolerances for Concrete Construction and Materials (ACI 117-10) and Commentary (ACI 117R-10)	ACI 117
Building Code Requirements for Structural Concrete	ACI 318-14
Specification for Hot Weather Concreting (2006) / Guide to Hot Weather Concreting	ACI 305/305R
Standard Specification for Cold Weather Concreting / Guide to Cold Weather Concreting	ACI 306/306R
Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary	ACI 355.2
Recommended Practice for Machinery Installation and Installation Design, 2nd Edition	API RP 686
Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel	ASTM B695
Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or [50-mm] Cube Specimens)	ASTM C109
Standard Test Method for Linear Shrinkage and Coefficient of Thermal Expansion of Chemical-Resistant Mortars, Grouts, Monolithic Surfacings, and Polymer Concretes	ASTM C531-12
Standard Test Methods for Compressive Strength of Chemical-Resistant Mortars, Grouts, Monolithic Surfacings and Polymer Concretes	ASTM C579-12
Standard Test Method for Bond Strength of Epoxy-Resin Systems Used with Concrete by Slant Shear	ASTM C882/C882M REV A

Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)	ASTM C1107/C1107M REV A
Standard Test Methods for Compressive Creep of Chemical-Resistant Polymer Machinery Grouts	ASTM C1181-12
Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners	ASTM F2329/F2329M

INTERNATIONAL STANDARDS

Quality management systems - Fundamentals and vocabulary	ISO 9000

7.3 PROJECT STANDARDS

1. The project and Engineering Contractor shall apply the standards as been specified above and no project specific technical standards and/or specifications shall be made on top of these.

It is recognized that the Engineering Contractor has its own procedures and standards to manage a project, from that perspective it is allowed to develop a limited number of project specific standards, specifications, procedures that are required to manage and administer the project execution.

APPENDIX 1 MIN SLAB THICKNESS FOR RIGID BEHAVIOR

Figure 1 Minimum slab thickness in m for rigid behaviour

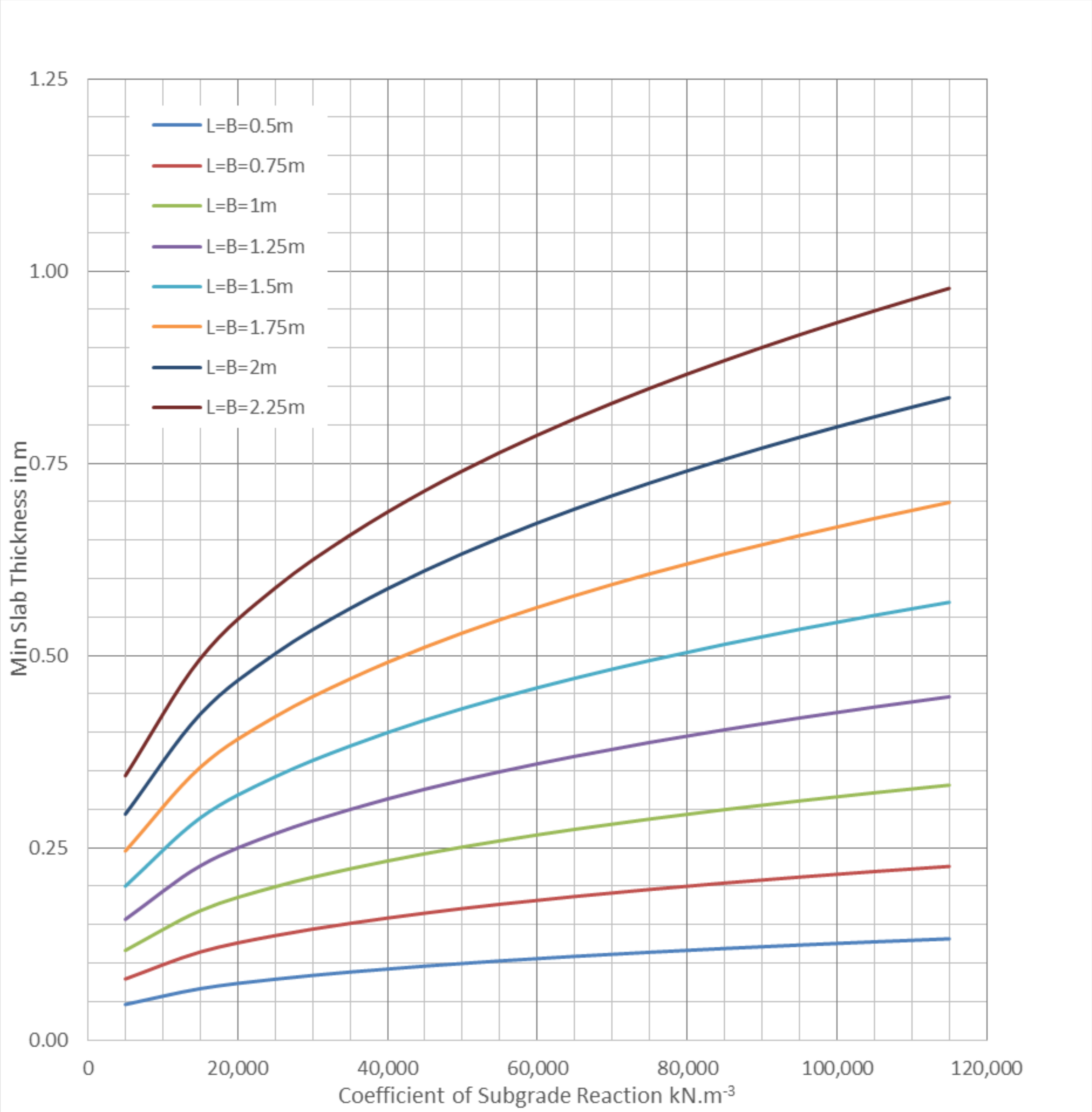

Figure 2 Minimum slab thickness in inches for rigid behaviour